

Distributions and Foliations

1.1 Distributions and Involutivity

Definition 1.1.1

Let M be a smooth manifold.

- A **distribution on M of rank k** is a rank- k subbundle of TM . It is called a **smooth distribution** if it is a smooth subbundle.
- Often a rank- k distribution is described by specifying for each $p \in M$ a linear subspace $D_p \subseteq T_p M$ of dimension k , and letting $D = \bigcup_{p \in M} D_p$. It then follows from the local frame criterion for subbundles (Lemma 10.32) that D is a smooth distribution if and only if each point of M has a neighborhood U on which there are smooth vector fields $X_1, \dots, X_k : U \rightarrow TM$ such that $X_1|_q, \dots, X_k|_q$ form a basis for D_q at each $q \in U$. In this case, we say that D is the distribution (locally) **spanned by the vector fields $X_1, \dots, X_k$** .

1.1.1 Integral Manifolds and Involutivity

Definition 1.1.2

Suppose $D \subseteq TM$ is a smooth distribution. A nonempty immersed submanifold $N \subseteq M$ is called an **integral manifold of D** if $T_p N = D_p$ at each point $p \in N$.

Definition 1.1.3

Suppose D is a smooth distribution on M . We say that D is **involutive** if given any pair of smooth local sections of D ^a, their Lie bracket is also a local section of D .

^a(i.e., smooth vector fields X, Y defined on an open subset of M such that $X_p, Y_p \in D_p$ for each p)

1.2 The Frobenius Theorem

Definition 1.2.1

Given a rank- k distribution $D \subseteq TM$, let us say that a smooth coordinate chart (U, φ) on M is **flat for D** if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and at points of U , D is spanned by the first k coordinate vector fields $\partial/\partial x^1, \dots, \partial/\partial x^k$.

Definition 1.2.2

We say that a distribution $D \subseteq TM$ is **completely integrable** if there exists a flat chart for D in a neighborhood of each point of M . Obviously, every completely

Theorem 1.2.3 ▶ Frobenius

Every **involutive distribution** is **completely integrable**.

1.3 Foliations

Definition 1.3.1

Let M be a smooth n -manifold, and let $\mathcal{F}$ be any collection of k -dimensional submanifolds of M .

- A smooth chart (U, φ) for M is said to be **flat for $\mathcal{F}$** if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and each submanifold in $\mathcal{F}$ intersects U in either the empty set or a countable union of k -dimensional slices of the form $x^{k+1} = c^{k+1}, \dots, x^n = c^n$.
- We define a **foliation of dimension k on M** to be a collection $\mathcal{F}$ of disjoint, connected, nonempty, immersed k -dimensional submanifolds of M (called the **leaves of the foliation**), whose union is M , and such that in a neighborhood of each point $p \in M$ there exists a flat chart for $\mathcal{F}$.